Genetic issues in the 21st century

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Many of you will be familiar with the term *genethics*, or the ethics (morals) behind genetic research. Since 2001, a few new terms have arisen, two of which you will probably hear at some time. These are *genomics*, a word that refers to an organism's genome (complete genetic make-up) and *proteomics*, which refers to the complete set of proteins found within an organism – and there are many!

It is just as important to understand and map the human proteome as it was to map the human genome – some researchers might argue it is more important. At one time, people believed that each protein was responsible for one, possibly two, tasks within the body. We now know that a single protein can be involved in over 1000 tasks. This is of particular concern to pharmaceutical researchers. Many current pharmaceuticals are designed to target and affect one or two proteins at most. Knowing that proteins work collectively, and in complex ways, revealed the importance of re-thinking current methods of pharmaceutical development.

1 Define *pharmaceuticals*.

Barbara McClintock: Founder of cytogenetics

American scientist Barbara McClintock (1902–1992) provided indisputable, physical evidence that genes are transposable. Her work proved that genetic material was exchanged between matching chromosomes and the shuffling of genes produced the variation we recognise between related organisms. Barbara McClintock, a student of Thomas Hunt Morgan, is reported to have endured considerable ridicule from people who thought she was wrong. In the 1930s, with the assistance of Harriet Creighton, she set about providing evidence for her theory. They used maize to show how genetic recombination occurs when chromosomes cross over each other, break and recombine their genetic material to produce new combinations. For her research, Barbara McClintock was awarded the 1983 Nobel Prize for Medicine and Physiology. She was the first woman in America to win an unshared Nobel Prize.

2 Define *transposable*.

Implications of gene technology

Gene technology is not new but its implications continue to generate debate. Genetically modified (GM) foods worry many people because there has not been time to assess the long-term effects of ingesting them. Some of the main aims of using gene technology include:

- *improving foods (plant or animal)*: pest and disease resistance, addition of vitamins and other substances currently deficient in millions of diets, increasing growth time (and in some cases size).
- *producing medicines in new ways:* for example, modifying certain crops to produce substances they did not produce before, such as developing potato crops to produce a protein vital to create an oral hepatitis B vaccine.
- *treating diseases resulting from defective genes:* for example, replacing defective genes with functional ones (a hybridisation technique).

3 What is meant by an *oral* vaccine?

4 Give one reason why some people are against GM foods.

5 Give one reason why some people are for GM foods.

Give definitions for the following terms:

6 genethics

7 genomics

8 proteomics

9 hybridisation

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At present *selective breeding* is used to produce plant and animal crops with desired features. Selective breeding directs the 'evolution' of a species by using only organisms that possess the desired trait(s) as breeding stock. In some instances, crossbreeding is used to do this. *Crossbreeding* uses closely related organisms that may not breed under natural circumstances but are still considered 'acceptably related' for the purpose.

Where gene technology differs drastically from this is that it can introduce genetic material from completely unrelated species. The 'mix and match' is made when researchers believe that an organism, if it possessed the ability to produce a given substance, could use this modification in their own genetic make-up to produce a new result.

How is the DNA modified?

Scientists identify and isolate the strand of DNA on which a required trait is found. The DNA is then replicated many thousands of times producing clones of the original strand. The area in which the required genes are located is identified and the required piece is spliced from the cloned strand. The spliced piece of DNA is inserted into the DNA of another organism. DNA is responsible for protein development and proteins instruct certain reactions to occur. Thus the introduced DNA will produce changes in the reactions that occur in the future.

This technology has been used to instruct the genes of ewes to produce substances in their milk, which will be extracted for the treatment of diseases such as cystic fibrosis. In some instances, gene therapy will be used to splice defective genes from an individual where it will be replaced by properly functioning genes (eg muscular dystrophy). Since it is impossible to replace every defective gene in a body, particularly in adults, some therapies will be used to clone many thousands of 'repaired' DNA, which will then be reintroduced into the individual, providing enough effective coding to correct (at least partially) the problem. Hopefully, in the case of developing embryos, for some conditions, defective DNA can be repaired or replaced before the child is born.

Give definitions for the following terms from the information above:

10 selective breeding

11 crossbreeding

12 isolate

13 replicated

14 spliced

15 ewes

16 therapies

17 Give one example of *selective breeding*.

18 Give one example of *crossbreeding*.

Research

19 There will always be a place for *traditional* and *organic* farming methods.
Write brief definitions for each method. Include an explanation of their similarities and differences.

20 In the box provided below, sketch the basic steps involved in DNA modification.

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